Series 1 Survival Analysis

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Exercise 1

A)Having the hazard funtcion $h(t) = \frac{\alpha}{\beta+t}$,$t > 0$ with $\alpha, \beta > 0$, it is simple to calculate the survival function $S(t) = e^{-\int_0^t h(t)\,dt}$, and the $f(t) = h(t)S(t)$.

i) Thus $-\int_0^t h(t)dt = -\int_0^t \frac{\alpha}{\beta+t}dt = -\alpha\int_0^t (ln(\beta+t))'dt = \alpha ln(\frac{\beta+t}{\beta})$, so $S(t) = e^{\alpha ln(\frac{\beta+t}{\beta})}$

or $S(t) = (\frac{\beta}{\beta+t})^\alpha$ and $f(t) = \frac{\alpha\beta^a}{(\beta+t)^{\alpha+1}}$.

ii) Now we want to calculate the left truncated survival function $S^*(t)$,

by definition $S^*(t) = P(T > t_0+t|T > t_0)$. But $S^*(t) = P(T > t_0+t|T > t_0) = \frac{S(t_0+t)}{S(t_0)} = \frac{(\frac{\beta}{\beta+t+t_0})^\alpha}{(\frac{\beta}{\beta+t_0})^\alpha}$. Or $S^*(t) = (\frac{\beta+t_0}{\beta+t_0+t})^\alpha$.

This too is a Pareto survival function with a new scale parameter $\beta' = \beta + t_0$ and the same α.

iii) Now we want to calculate the mean residual life $m(t_0)$ and the conditional expectation $E[T|T > t_0]$.

To do that we will use the left truncated survival function we just computed:

$m(t_0) = E[T-t_0|T > T_0] = \int_0^\infty S^*(t)dt = \int_0^\infty (\frac{\beta+t_0}{\beta+t_0+t})^\alpha dt = \int_0^\infty (\frac{\beta'}{\beta'+t})^\alpha dt$ for $\beta' = \beta+t_0$.
We continue $m(t_0) = \beta'^\alpha \int_{\beta'}^\infty (\frac{1}{u})^\alpha du = \beta'^\alpha[\frac{1}{\alpha-1}(\frac{1}{u})^{\alpha-1}]_{\beta'}^\infty$.

Finally $m(t_0) = \frac{\beta'}{\alpha-1} = \frac{\beta+t_0}{\alpha-1}$

From that we simply calculate the conditional expectation:

$E[T|T > t_0] = t_0 + m(t_0) = \frac{\beta+\alpha t_0}{\alpha-1}$

B) Using the Pareto with $\alpha = 4, \beta = 5$, assuming that's the distribution the lifetime of a battery follows, we can calculate the following:

i) To calculate what is the probability for the battery to survive at least a)2 and b)10 years:

```r
# Parameters
alpha <- 4
beta  <- 5

# Survival function S(t)
S <- function(t) (beta / (beta + t))^alpha

# Survival probabilities
P2 <- S(2)
P10 <- S(10)

cat("P(T > 2)  =", P2, "\n")
```

```
P(T > 2)  = 0.2603082
```

```r
cat("P(T > 10) =", P10, "\n\n")
```

```
P(T > 10) = 0.01234568
```

Or the probability of the battery to survive more than 2 years is around 26% and for more than 10 years is 1.2%.

ii) We can calcuate the mean and the median:

```r
# Density f(t)
f <- function(t) alpha * beta^alpha / (beta + t)^(alpha + 1)

# Mean lifetime
E_T <- beta / (alpha - 1)

# Median
median_T <- beta * (2^(1/alpha) - 1)

cat("E[T] =", E_T, "\n")
```

```
E[T] = 1.666667
```

```r
cat("Median =", median_T, "\n\n")
```

```
Median = 0.9460356
```

Or the mean lifetime of the battery is roughly 1.667 years and the median is 0.946 years.

iii) For $t_0 = 1$ year we can calculate $m(1)$ and the conditional $E[T|T > 1]$:

```
# Mean residual life m(t0)
m <- function(t0) (beta + t0) / (alpha - 1)

# Conditional expectation E[T | T > t0]
E_cond <- function(t0) t0 + m(t0)

t0 <- 1
m_t0 <- m(t0)
E_cond_t0 <- E_cond(t0)

cat("m(1) =", m_t0, "\n")
```

```
m(1) = 2
```

```
cat("E[T | T > 1] =", E_cond_t0, "\n\n")
```

```
E[T | T > 1] = 3
```

Thus the expected lifetime remaining at the age of 1 years is $m(1) = 2$ years and the conditional expectation $E[T|T > 1] = 3$ years. iv) We can also calculate at what time t_0 must a battery be replaced if we want 99% of batteries to still be operational or what is the t_0 such that $P(T > t0) = 0.99$.

```
target <- 0.99
t0_99 <- beta * (target^(-1/alpha) - 1)

cat("t0 such that P(T > t0) = 0.99:", t0_99, "\n")
```

```
t0 such that P(T > t0) = 0.99: 0.01257872
```

Turns out that $t_0 = 0.0125$ years or 4.6 days.

Exercise 2

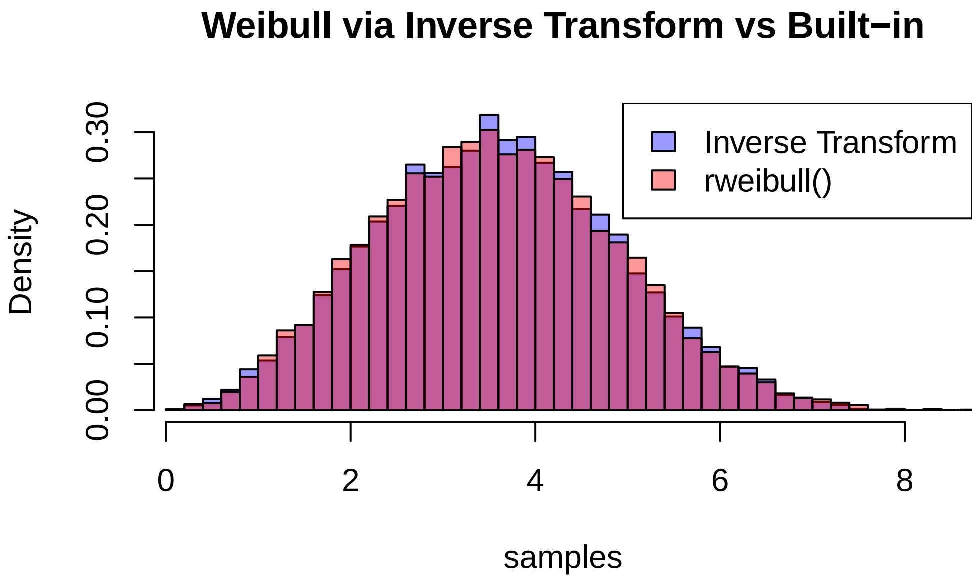
If $Y = F(X) \sim U[0,1]$ then if I assume that Y is the Weinbull distribution then it to must follow a uniform distribution, so if I can inverse the function and express X as a function of Y then I can create the distribution by using a uniform as a base $X = F^{-1}(Y)$:

$$F(X) = 1 - e^{-(X/\lambda)^k} \Rightarrow Y = 1 - e^{(X/\lambda)^k} \Rightarrow -(X/\lambda)^k = ln(1-Y) \Rightarrow X = \lambda[-ln(1-Y)]^{1/k}$$

Where $Y \sim U(0,1)$.

To simulate this properly we construct the following script that generates the distribution with the inverse transform method and compares it to the built in one of R itself:

```
# Weibull parameters
k <- 3      # shape
lambda <- 4  # scale

# Inverse transform generator
rweibull_inv <- function(n, k, lambda) {
  U <- runif(n)
  lambda * (-log(U))^(1/k)
}

# Generate 10,000 samples
set.seed(444)
samples <- rweibull_inv(10000, k, lambda)

# Compare with built-in rweibull()
samples_builtin <- rweibull(10000, shape = k, scale = lambda)

# Plot comparison
hist(samples, breaks = 40, probability = TRUE,
     col = rgb(0,0,1,0.4), main = "Weibull via Inverse Transform vs Built-in")
hist(samples_builtin, breaks = 40, probability = TRUE,
     col = rgb(1,0,0,0.4), add = TRUE)

legend("topright",
       legend = c("Inverse Transform", "rweibull()"),
       fill = c(rgb(0,0,1,0.4), rgb(1,0,0,0.4)))
```



It is clear that the generated distribution is very close to the built in one of R, thus this is a very solid method of creating distributions based on the uniform. This is a very useful result for simulations and MCMC and is used in a plethora of fields.